Tropical Cyclone
Spatial Analysis Report

TABLE OF CONTENTS

Introduction	**3**
Objective	**3**
Methodology	**3**
Analysis	**5**
Discussion	**7**
Limitations	**8**
Conclusion	**8**
Glossary of Terms	**9**

Introduction

Tropical cyclones are among the most destructive atmospheric systems, capable of generating extreme winds, storm surge, and widespread environmental impacts. This report applies geospatial analysis techniques to examine tropical cyclone patterns, trajectories, and areas of influence within the study region. Using GIS Tools, the analysis identifies key spatial trends, evaluates proximity-based impacts, and highlights locations most frequently affected.

Objective

The goal of this study is to identify where tropical cyclones commonly occur and determine which cities fall within their areas of influence using GIS-based spatial analysis. Understanding their spatial behaviour is essential for assessing risk, improving early-warning systems, and supporting evidence-based disaster management.

Methodology

The analysis was conducted using a cyclone dataset (from IBTrACS) I which contained every storm reliably recorded. IBTrACS - International Best Track Archive for Climate Stewardship

1. Data Sources

- **Cyclone Data (Point Features):**
  Cyclone track data was obtained from the International Best Track Archive for Climate Stewardship (IBTrACS). The dataset provides historical cyclone paths worldwide, including key attributes such as storm name, track coordinates (latitude and longitude), wind speed, pressure and storm category. To restrict the analysis to tropical storms and stronger systems, the data was filtered in QGIS using an attribute-based selection-records with a wind_ssh value > 0 were retained, as this indicates systems that reached at least tropical storm intensity on the Saffir-Simpson scale, excluding weaker or non-classified systems. I used the dataset's USA_SSHS (United States Saffir–Simpson Hurricane Scale) for the heatmap intensity.

- **Base Map Layer (Raster Imagery):**
  A high resolution **ESRI Satellite** imagery layer was used to provide essential geographic context for visualising the fire locations. This layer offers a detailed visual reference of land cover and topography, aiding in

the interpretation of fire locations relative to built-up areas, vegetation, and major geographic features. The layer was accessed via the Web QuickMapServices plugin in QGIS.

- **Storm Buffer Area (Polygon Layers):**
  A buffer zone was generated around each cyclone track using the **Buffer tool** in QGIS. A fixed buffer distance of 100 km was applied to represent the estimated zone of cyclone impact, accounting for strong winds, heavy rainfall, and storm surge effects from the storm centre.

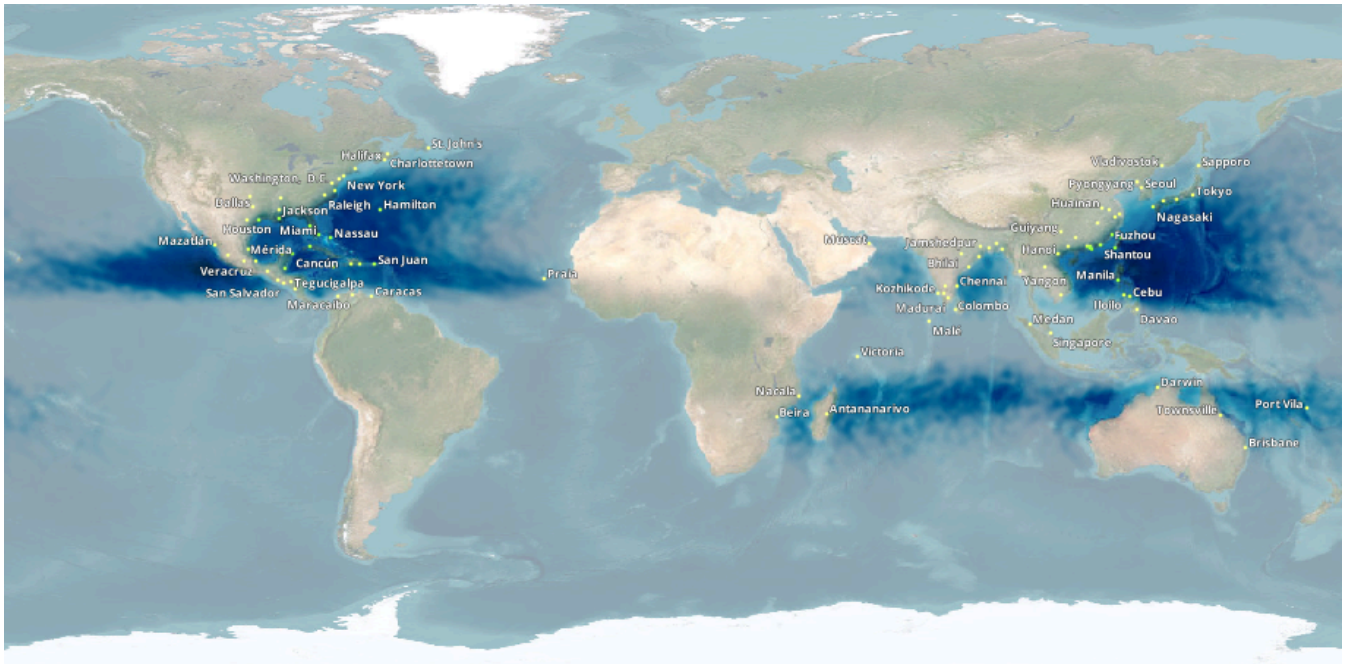
- **Relevant Cities (Point Features):**
  A global populated-places dataset from Natural Earth (10m Populated Places) was used to identify major cities potentially affected by cyclone activity. This data was filtered in QGIS using the expression ("SCALERANK" <= 3) to retain only major cities with higher regional significance for clearer spatial interpretation. After that, the relevant cities were extracted by spatially **filtering** the complete cities point dataset using the cyclone buffer polygons. This was achieved in QGIS using a spatial intersection, retaining only city features that intersected the storm buffer zones.

2. Symbology and Visualisation

The cyclone heatmap consisted of around 249k points, each point representing a recorded position of a tropical cyclone at a given time. Point intensity was weighted using the **USA_SSHS** attribute, which classifies cyclone strength based on the Saffir-Simpson Hurricane Scale derived from maximum sustained wind speed.

A **white-to-dark-blue colour ramp** was applied to visualise spatial variations in cyclone intensity. Lighter colours indicate lower-intensity systems (tropical depressions and tropical storms, while progressively darker blues represent higher-intensity cyclones (Category 1-5).

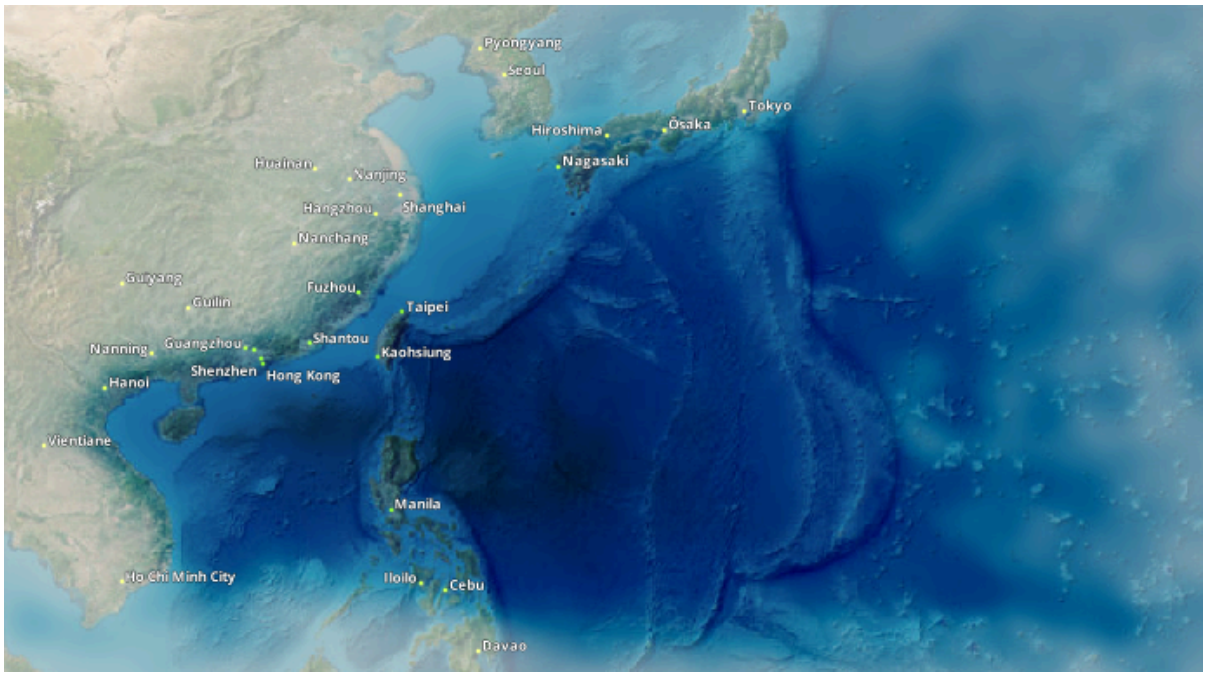
Analysis



1. North Atlantic & Eastern Pacific Basin

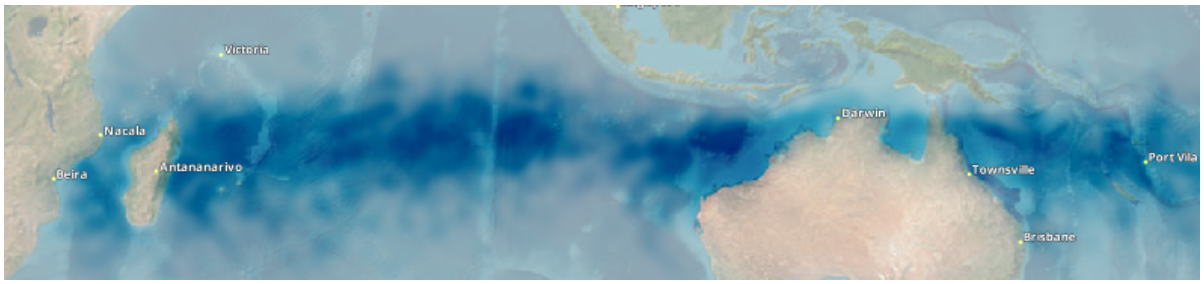
This basin shows a high concentration of tropical cyclones forming over the warm waters of the Caribbean Sea and the eastern Pacific Ocean. Cyclone tracks frequently move westward and northwestward, impacting Central America, the Caribbean islands, and eastern coastline of Mexico. The spatial pattern indicates strong clustering near coastal regions, with cyclone activity decreasing rapidly inland, highlighting the dependence of cyclone systems on warm ocean waters.

2. Western North Pacific Basin



This basin exhibits the highest density of tropical cyclones among the identified regions. Cyclones commonly form east of the Philippines and track northwest toward East Asia, frequently affecting the Philippines, Taiwan, and Japan. The wide spatial spread of cyclone paths suggests high variability in cyclone movement, while repeated overlap of tracks indicates this region as a persistent cyclone hotspot.

3. South Indian Ocean & South Pacific Basin



Cyclone activity in the southern hemisphere is concentrated within the South Indian Ocean and South Pacific basins. In the South Indian Ocean cyclones frequently impact Madagascar and the eastern coast of southern Africa. Further east, cyclone tracks are observed across the South Pacific, affecting island nations such as Fiji, Vanuatu, and Tonga. These cyclones predominantly remain over oceanic regions and tend to weaken as they approach land, particularly over larger landmasses.

Discussion

The spatial analysis of global tropical cyclone tracks reveals several clear and physically consistent **patterns** across major ocean basins.

In the **Indian ocean**, cyclone movement generally follows an **east-to-west trajectory**, with storms often forming near or north-west of Australia and progressing toward the western Indian Ocean, occasionally impacting regions such as Madagascar and eastern Africa. This pattern is strongly influenced by prevailing trade winds and large-scale atmospheric circulation, which steer cyclones westward once they form.

In the **Pacific Ocean**, cyclone tracks appear **curved** rather than linear. This circular or arcing movement is not random; it is primarily caused by the **Coriolis effect**, which results from Earth's rotation and causes storms to curve as they move away from the equator. While this motion may visually resemble a "loop" or vortex-like path, it is a **predictable physical process** rather than an anomalous phenomenon.

Cyclone behaviour in the **Atlantic** is less immediately distinct in comparison. Atlantic hurricanes typically form off the west coast of Africa and move westward toward the Caribbean and the eastern United States, before curving northward or northeastward into the Atlantic Ocean. The variability in these tracks can make the overall pattern appear less uniform when visualised globally.

Across all regions, the **Pacific Ocean** stands out as the **dominant** source of tropical cyclones. This aligns with climatological expectations, as the Pacific is the largest ocean basin and contains extensive areas of **warm** sea surface temperatures that provide the energy required for cyclone formation.

Additionally, smaller and more **isolated** storm systems are visible near **India and the northern Indian Ocean**. These systems **do not conform** to the 3 major cyclone track patterns and are likely influenced by regional factors such as monsoon dynamics, land-sea temperature contrasts, and shorter storm lifespans.

Finally, cyclone formation is largely concentrated near the **equator**, but not directly on it. Most storms form approximately between 5° and 20° latitude north or south, where sea surface temperatures are sufficiently warm and the Coriolis effect is strong enough to initiate and sustain cyclonic rotation.

Limitations

Several limitations should be considered when interpreting the observed cyclone patterns:

1. The analysis is limited by the temporal scope of the dataset. Cyclone behaviour **varies significantly** between years due to climate oscillations such as El Niño and La Niña. As a result, the identified patterns may not fully represent **long-term** global cyclone trends.

2. The study relies on **filtered** cyclone intensity data, which excludes weaker or short-lived systems. While this improves clarity by focusing on impactful storms, it may **underrepresent** the total frequency of cyclone formation, particularly in regions where storms **dissipate quickly**.

3. The use of track-based point data simplifies cyclone movement into discrete locations rather than continuous systems. This may exaggerate curvature or clustering effects, especially in the Pacific Ocean where cyclone paths appear circular when visualised at a global scale.

4. **Regional variation in data quality** may influence results. Cyclone detection and reporting are generally **more accurate** in **heavily monitored** regions such as the Pacific and Atlantic, while smaller storms in the Indian Ocean or near developing regions may be underreported.

5. The analysis is **spatially constrained** to **latitude-based** formation zones near the equator. This limits conclusions about cyclone behaviour at higher latitudes where storms often transition into **extratropical systems** and are no longer classified as tropical cyclones.

Conclusion

Overall, cyclone activity is **concentrated** near the **equator**, where atmospheric and oceanic conditions are **most favourable** for development, though formation does not occur directly on the equatorial line. Despite data and methodological limitations, the analysis demonstrates that global cyclone behavior follows **consistent** and physically driven spatial patterns.

These findings highlight the value of GIS-based approaches in visualising and interpreting large-scale atmospheric phenomena.

Glossary of Terms

Cyclone - A large-scale storm system characterised by strong winds rotating around a low-pressure center.

Tropical Storm - A type of cyclone with sustained winds between 63-118km/h.

Hurricane / Typhoon / Cyclone (regional names) -
Different names for strong tropical storms depending on the region.
Hurricane: Atlantic/Eastern Pacific | **Typhoon**: Northwest Pacific
| **Cyclone**: Indian Ocean/South Pacific

Saffir-Simpson Hurricane Scale (SSHS) - A system used to classify the intensity of hurricanes from Category 1 (weakest) to Category 5 (strongest) based on wind speed.

Buffer Zone - An area created around a feature (like a cyclone path) to analyze surrounding elements, e.g., cities within a certain distance from a storm.

Filtering - The process of selecting only certain data points that meet specific criteria (e.g., only storms with wind speed above 0 km/h.).

Point Features - Specific locations represented as points on a map in GIS software.

Intensity - The strength or severity of a cyclone, often measured by wind speed.

Low-Pressure Center - The area at the heart of a cyclone where air pressure is lower than the surrounding areas, causing air to flow inward and rotate.

Wind Speed (km/h) - The rate at which air is moving in a cyclone; used to determine the storm's category.

Heatmap - A visual representation of data where values are shown using colours, often used to show intensity or frequency of events.

Tropical Cyclone Path / Track - The route taken by a cyclone as it moves over the ocean or land.